Fractional-Slot Concentrated-Windings Synchronous Permanent Magnet Machines: Opportunities and Challenges

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***Abstract*—Fractional-slot concentrated-winding (FSCW) synchronous permanent magnet (PM) machines have been gaining interest over the last few years. This is mainly due to the several advantages that this type of windings provides. These include high-power density, high efficiency, short end turns, high slot fill factor particularly when coupled with segmented stator structures, low cogging torque, flux-weakening capability, and fault tolerance. This paper is going to provide a thorough analysis of FSCW synchronous PM machines in terms of opportunities and challenges. This paper will cover the theory and design of FSCW synchronous PM machines, achieving high-power density, flux-weakening capability, comparison of single- versus double-layer windings, fault-tolerance rotor losses, parasitic effects, comparison of interior versus surface PM machines, and various types of machines. This paper will also provide a summary of the commercial applications that involve FSCW synchronous PM machines.**

***Index Terms*—Concentrated, distributed, fractional slot, generators, integral slot, machines, motors, permanent magnet (PM), synchronous, windings.**

I. Introduction

THE WINDING configurations which are most commonly employed for three-phase radial-field permanent magnet (PM) brushless machines, can be classified as [1]:

1) overlapping, either distributed [Fig. 1(a)] (two slots/pole/phase) or concentrated [Fig. 1(b)] (one slot/pole/phase);
2) nonoverlapping, i.e., concentrated, with either all teeth wound [Fig. 1(c)] or alternate teeth wound [Fig. 1(d)].

Nonoverlapping windings will be referred to as fractional-slot concentrated-winding (FSCW) for the rest of this paper. All teeth wound winding will be referred to as double-layer (DL), while alternate teeth wound will be referred to as single-layer (SL). Fig. 2(a) and (b) shows actual prototypes of both types of windings [2].

Since a distributed overlapping winding generally results in a more sinusoidal magnetomotive force (MMF) distribution and EMF waveform, it is used extensively in PM brushless ac (BLAC) machines.

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On the other hand, FSCW synchronous PM machines have been gaining interest over the last few years. This is mainly due to the several advantages that this type of windings provides. These include high-power density, high efficiency, short end turns [3], [4], high slot fill factor particularly when coupled with segmented stator structures, low cogging torque, flux-weakening capability, and fault tolerance. Table I summarizes the key differences between distributed windings and FSCW.

This paper is going to provide a thorough analysis of FSCW synchronous PM machines in terms of opportunities and challenges. This paper will cover the key areas and publications related to FSCW.

Over the last few years, as FSCW started gaining many attention, one of the key challenges was to understand the theory of operation of FSCW as well as how to systematically layout the windings to achieve maximum winding factor. Section II will cover the key publications that addressed the theory behind using FSCW and how to achieve an optimal winding layout for any slot/pole combination in addition to other improved performance aspects.

The design and analysis of FSCW PM machines is challenging in the sense that the winding configuration deviates significantly for the conventional sinusoidal distribution. Section III will cover the key publications addressing the design and analysis of FSCW PM machines.

One of the key advantages of FSCW is the ability to achieve significantly higher copper slot fill factor (compared to conventional laminated stator structures) if coupled with segmented stator structures. This can have a significant impact on the machine power density. Section IV will cover the key papers addressing various concepts of segmenting the stator structure to significantly increase the copper slot fill factor.

Traditionally, surface PM (SPM) machines have the reputation of poor flux-weakening capability. One of the key advantages of FSCW is that they help achieve a wide speed range of constant power operation. Section V will cover the key papers addressing the flux-weakening capability of SPM machines equipped with FSCW.

Section VI will cover the key papers comparing the various types of FSCW mainly SL versus DL winding configurations. The key differences and tradeoffs will be highlighted.

One of the key challenges of using FSCW configurations is the significant rotor losses (including magnet losses, rotor core losses, and sleeve losses in case of conductive sleeve) particularly at high speeds due to the various sub- and

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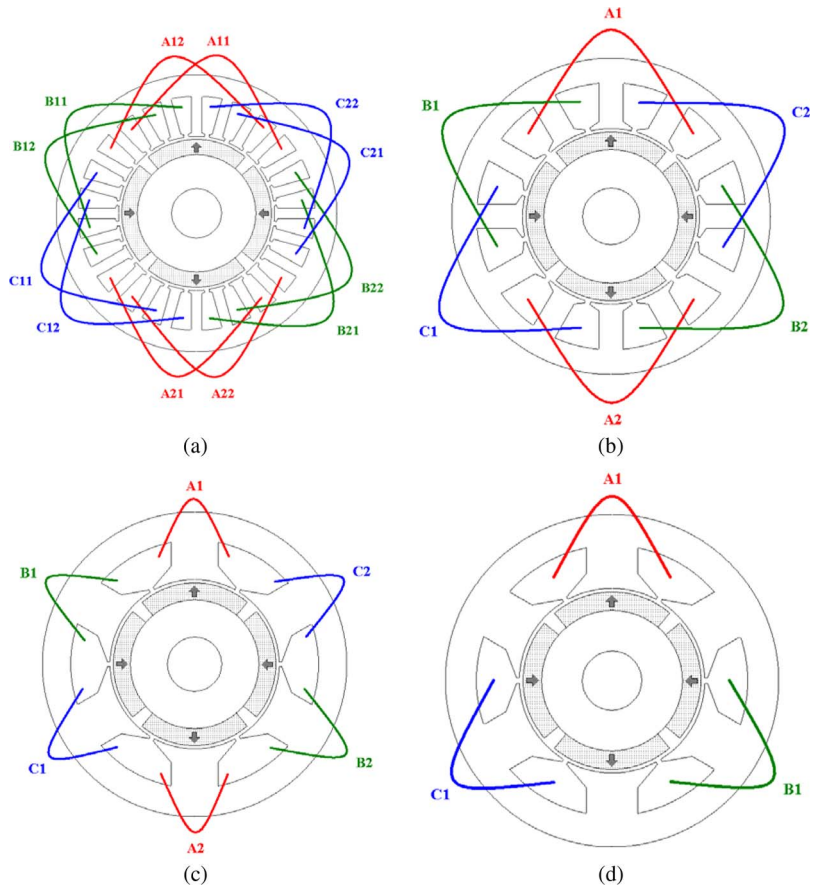

Fig. 1. Typical stator winding configurations (four pole) [1]. (a) Twenty-four slot, overlapping (distributed). (b) Twelve slot, overlapping (concentrated). (c) Six slot, nonoverlapping, all teeth wound. (d) Six slot, nonoverlapping, alternate teeth wound.

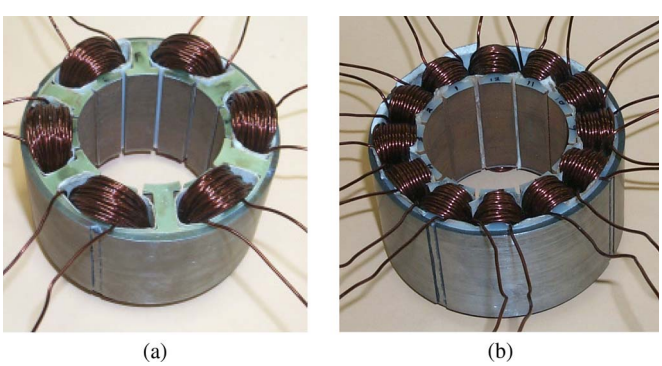

Fig. 2. (a) Twelve-slot/ten-pole design with SL winding [2]. (b) Twelve slot/ten pole with DL winding [2].

TABLE I
COMPARISON OF DISTRIBUTED AND CONCENTRATED WINDINGS

	Distributed Windings	Concentrated Windings
Typical copper slot fill factor	35%-45%	50%-65% (if coupled with segmented stator structures)
Stator structure	Continuous laminations	Continuous laminations or segmented structures
End turns	Long overlapping	Short non-overlapping
Torque-producing stator space harmonic component	Fundamental	In most cases (except for 0.5 slot/pole/phase) a higher order harmonic

superspace-harmonic components inherent to such winding configurations that are not in synchronism with the rotor. Section VII will cover the key papers addressing the various aspects of rotor losses in PM synchronous machines using FSCW.

Parasitic effects such as noise, vibration, unbalanced magnetic forces, and torque ripple are always a concern when designing an electrical machines. These parasitic effects can potentially be higher in FSCW PM machines due to the additional harmonic contents. Section VIII will cover the key papers addressing parasitic effects in FSCW PM machines.

Fault tolerance is one of the key issues with PM machines in general, particularly in safety-critical applications. The main reason is that the PMs cannot be deexcited in case of a fault particularly in case of a generator that is coupled to a prime mover. FSCW provide many advantages in terms of fault tolerance particularly SL windings. SL windings provide very low mutual coupling between the various phases as well as physical separation. Section IX will cover the key papers addressing fault tolerance in PM synchronous machines using FSCW configurations.

Even though most of the work done up to date focused on SPM machines, there is a growing interest in the use of interior PM (IPM) machines equipped with FSCW. The hope is that FSCW IPM machines will combine the benefits of the FSCW previously mentioned in addition to the benefits of an IPM rotor in terms of potentially reducing magnet content as well as easier magnet retention compared to SPM machines. This area is by no means mature, and much work is needed to fully

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understand the tradeoffs involved. Section X will focus on the key papers that provided comparison between IPM and SPM machines with FSCW configurations.

The main focus so far has been on radial-flux FSCW PM machines. There is a growing interest in other types of PM machines equipped with FSCW. Section XI will cover the key papers addressing the use of FSCW in other types of machines mainly axial-flux, tubular PM machines, and flux-switching machines.

Section XII will also provide a summary of the commercial applications that involve FSCW PM machines as well as the potential future evolution of FSCW in new applications and fields.

II. Theory

Over the last few years, as FSCW started gaining attention, one of the key challenges was to understand the theory of operation of FSCW as well as how to determine the slot/pole combinations that can support FSCW and be able to systematically layout the windings to achieve maximum winding factor. Several authors proposed various methods to achieve this goal.

This section will cover the key publications that addressed the theory behind using FSCW and how to achieve an optimal winding layout for any slot/pole combination in addition to other improved performance aspects.

In [3], Cros and Viarouge presented an enlightening study about the use of concentrated windings in high-performance PM machines. They identified the various slot/pole combinations that can support three-phase concentrated windings. In addition, they presented a systematic method to determine the optimum concentrated winding layout in both cases of regular and irregular slot distribution. They provided some guidelines for identifying the slot/pole combinations that can provide high machine performance. They provided analysis results for sample designs using concentrated windings showing that the performance of these machines are better than that of traditional machines with one slot/pole/phase. There is minimization of both copper volume and Joule losses, reduction in the manufacturing cost and improvement in the output characteristics. The production process of these designs can be further simplified if coupled with the use of segmented soft magnetic composite (SMC) structures.

In [4], Magnussen and Sadarangani presented a method for calculating winding factors for electrical machines equipped with concentrated windings based on phasor relationships. The effect of the winding factor on the Joule losses has been discussed. A comparison of the Joule losses, cogging torque, and axial length of conventional distributed one slot/pole/phase winding, SL-concentrated winding, and DL-concentrated winding has been presented. It was shown that by choosing the appropriate slot/pole combination, concentrated windings have lower Joule losses and cogging torque compared to distributed windings. In addition, it was shown that the DL concentrated windings has the shortest axial length and hence has the greatest potential to be the most compact unit among the three winding configurations under consideration.

In [5], [6], Bianchi *et al.* presented another method to determine the optimal winding layout for the various slot/pole combinations as well as design FSCW PM machines based on the stator star of slots.

In [7], EL-Refaie *et al.* expanded the work that was presented in [3] to cover four-, five-, and six-phase configurations. Tables including the winding factors, cogging torque indicators, and net radial force indicators for the various slot/pole combinations have been provided. In addition to these three parameters, a key parameter for choosing the optimal slot/pole combination has been introduced. This parameter is the rotor loss figure of merit (FOM) that helps compare the rotor losses for the various slot/pole combinations on relative basis. The values for the rotor loss FOM have been evaluated for three-, four-, five-, and six-phase designs. Together, all these sets of tables would help a machine designer to converge to the optimal slot/pole combination and number of phases based on his application requirements.

In [8], Katsuma and Kitoh presented a brushless SPM motor with concentrated windings in the stator. They identified criteria for determining the number of rotor poles and the number of stator slots. Following this criteria, a design can be attained in which the minimum slot opening necessary for winding efficiently is ensured, the widths of the rotor poles and stator salient poles are equal, and m-phase windings can be attained without adverse effects upon the induced voltage waveforms.

In [9], Konecny presented a compact three-phase SPM machine using concentrated windings around the teeth. The torque ripple and vibrations are minimized while maximizing efficiency and starting torque per unit volume of the winding. He also presented formulas to determine the number of rotor poles and stator slots to achieve this. The effect of radial forces was overlooked in this paper.

In [10], Nishio presented a three-phase direct-current motor using concentrated windings around the teeth. They presented criteria for choosing the number of poles and the number of slots to minimize the cogging torque. The effect of radial forces was overlooked in this paper.

In [12], Huang and Hartman presented a high-speed outer-rotor SPM brushless dc (BLDC) motor. They highlighted the advantages of using the 12-slot/10-pole combination form the point of view of eliminating any net radial forces and cogging torque minimization.

In [13], Dhawan and Soghomonian discussed the design and characteristics of an outer rotor seven-phase brushless SPM motor using concentrated windings as well as the inverter electronics needed to control such a motor. It was shown that the inverter electronics are inherently smaller for this kind of motor compared to a three-phase brushless PM motor. The patents [13]–[17] discuss the various design aspects of this motor.

In [19], Libert and Soulard investigated various slot/pole combinations for SPM machines equipped with concentrated windings. Among the considered factors were the winding factors, MMF harmonic contact, torque ripple, and radial magnetic forces that cause vibration and noise.

In [20], Reichert discussed the advantages and disadvantages of using concentrated windings in large synchronous machines for low-speed high-torque applications. He indicated that the eddy current losses in the magnets might be a limiting factor for using concentrated windings in high-speed applications.

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In [21], Xia *et al.* presented the full analysis of a three-phase 24-slot/22-pole modular IPM machine for automotive applications. Modular PM machine is a relatively new topology [21] which is a subset of the concentrated winding topologies. In that case, all the coils, which belong to the same phase, are concentrated and wound either on consecutive or alternate teeth such that the phase windings do not overlap. It was shown that this machine has higher torque capability than conventional BLDC machine while the ripple and cogging torque are lower. In addition, the mechanical issues and eddy current losses induced in the magnets were taken into consideration.

In [23], Wang *et al.* identified the feasible slot/pole combinations for three-phase modular PM machines and their relative merits have been discussed. Analytical formulas for calculating the air-gap magnetic field have been presented. These can be used for calculating the back EMF, machine inductances, and the output torque.

In [24], Di Gerlando *et al.* presented a full performance analysis of a SPM machine equipped with a two-layer concentrated winding configuration. It was shown that this machine has very good back EMF and torque waveform quality as well as being capable of self-starting.

In [25], Noel *et al.* proposed some ideas and guidelines for the use of concentrated windings to reduce the short-circuit current in low-speed high-torque applications.

Various PM machine designs for various applications equipped with concentrated windings around the teeth are discussed in [25]–[31]. The details of these designs will not be discussed here but they are listed for the benefit of the reader.

III. Design and Analysis

The design and analysis of FSCW PM machines is challenging in the sense that the winding configuration deviates significantly for the conventional sinusoidal distribution. The standard $d-q$ analysis might not be applicable or accurate in the case of FSCW. This section will cover the key publications addressing the design and analysis of FSCW PM machines.

In [32], Bianchi *et al.* presented the most complete publication up to date summarizing most of the work that has been done regarding the theory and design of FSCW PM machines.

In [33], EL-Refaie *et al.* presented a closed form analytical method for the design and analysis of this type of machines. The proposed method is based on analytically calculating the magnetic field in the air gap and building upon this to calculate the various machine parameters and performance on a per-phase basis. This approach is suitable for nonsalient SPM machines but other analytical approaches need to be developed for salient IPM machines. The various machine performance aspects have been verified using finite-element analysis (FEA).

IV. High-Power Density

One of the key advantages of FSCW is the ability to achieve significantly higher copper slot fill factor (compared to conventional laminated stator structures) if coupled with segmented stator structures particularly if the windings are prepressed.

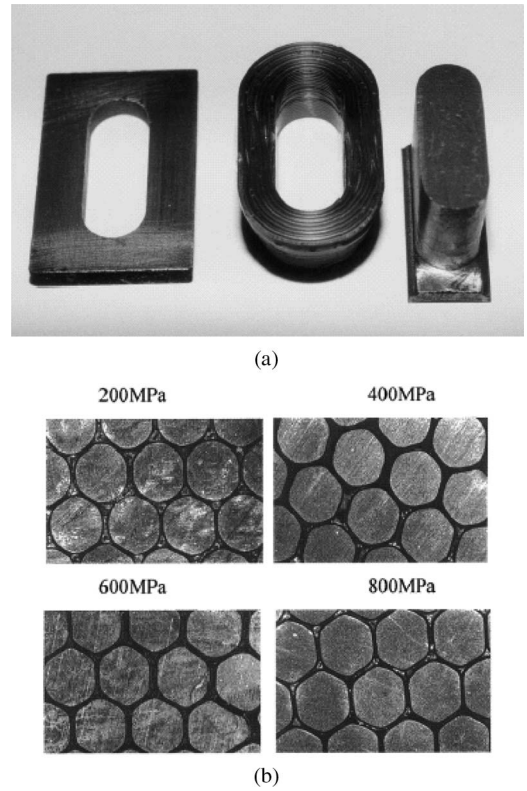

Fig. 3. (a) Manufactured core components and coil [11]. (b) Pressing trial results—coil sections [11] (78% fill factor reported).

This can have a significant impact on the machine power density. The copper slot fill factor is defined as

$$K_{\text{cu-fill}} = \frac{A_{\text{cu}}}{A_{\text{slot}}} \tag{1}$$

where A_{cu} is the total copper area and A_{slot} is the total slot area.

Several methods have been proposed to achieve this goal. This section will cover the key papers addressing various concepts of segmenting the stator structure to significantly increase the slot fill factor and reduce manufacturing cost.

In [11], Jack *et al.* reported a significantly high slot fill factor of $\sim$78% using SMCs and prepressed windings, as shown in Fig. 3.

Similar values of slot fill factor values have been reported in case of segmented laminated stator structures using plug-in-tooth technique [34]. Such configuration is shown in Fig. 4.

In [35], more recently, Akita *et al.* reported a 75% slot fill factor using a "joint-lapped core." Such a configuration is shown in Fig. 5. Table II summarizes the key differences between the three approaches.

There have been several publications addressing the use of FSCW in conjunction with segmented stator structures for various types of PM machines.

In [36], Cros *et al.* presented two prototypes of brushless PM motors using a dielectromagnetic material and concentrated windings, which can advantageously replace the conventional dc motor used as an automotive electric fan. The magnetic structure of the first prototype is equipped with small teeth localized between the main poles. Experimental results showed

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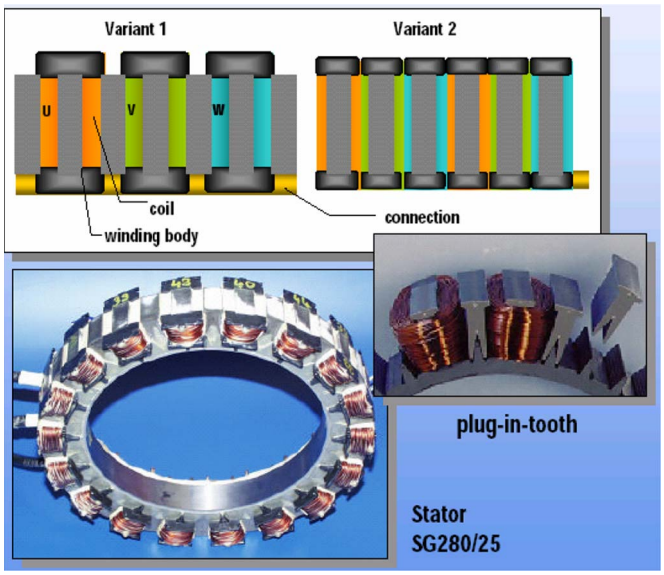
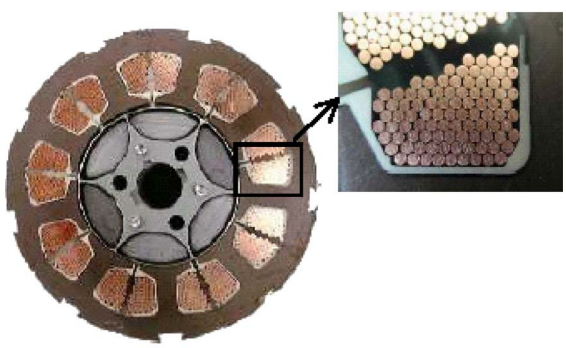
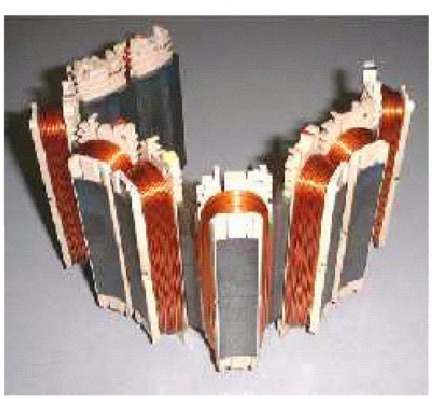


Fig. 4. Example of stator structure using laminated plug-in-tooth technique [34].



(a)



(b)

Fig. 5. (a) Cross section of a joint-lapped core machine [35] (75% fill factor reported). (b) Joint-lapped core after winding [35].

that the torque performances are the same as in the case of classical structures with distributed windings. In addition, there is a significant gain in the weight of copper ($> 50\%$) and in the total cost.

In [37], Cros *et al.* presented new structures of brush dc motor armatures with a plurality of simple coils wound around the same tooth and made of SMC. The performance in terms of torque-to-motor volume ratio and torque-to-copper volume ratio is higher than in the classical structures. Their performance is similar to that of the classical structures in terms of current commutation. Since these structures are well adapted to the use of SMC material and to a direct pressing process in a single process, the total production cost is minimized, and they can be used over a wide power range.

TABLE II
COMPARISON OF VARIOUS METHODS FOR SEGMENTING STATOR

	SMC	Plug-In Teeth	Joint-lapped core
Copper slot fill factor	78% with pre-pressed coils	60%-65	75%
Stator back iron rigidity	Lower rigidity due to segmented back iron	Rigid stator back iron	Lower rigidity due to segmented back iron
Number of pieces and tolerances	2X the number of stator teeth	1X the number of stator teeth	1X the number of stator teeth
Type of stator structure	Solid pieces of SMC	Laminated structures	Laminated structures
Saturation level	Lower saturation level of SMC	Same saturation level as in continuous structures	Same saturation level as in continuous structures

In [38], Cros *et al.* presented a new structure of the universal motor using SMC and concentrated windings. The stator core presents a claw-pole structure while the armature is equipped with concentrated winding with several coils wound around the same tooth. It was shown that with this new structure there is almost a 200% reduction in volume compared to a conventional universal motor structure with nearly the same performance.

In [39], Cros and Viarouge presented new structures of polyphase claw-pole machines. The whole concept depends on dividing the stator into three parts made of SMC and the use of centralized-concentrated winding with smaller number of coils. This simplifies the manufacturing process, mechanical assembly and winding realization. They also presented a design approach for these 3-D designs. They demonstrated that such 3-D designs could be derived form their equivalent 2-D designs with concentrated windings.

In [40], Cros *et al.* presented a comparison of different structures of PM BLDC motors with concentrated windings and segmented stator for low-power and high-efficiency application. Some structures have irregular slot distribution. It was possible to improve the overall performance compared to a commercial motor. The results were verified experimentally.

V. FLUX WEAKENING

Surface-mounted PM BLAC machines have often been considered to be poor candidates for achieving wide constant power operation by means of flux weakening. The principal reason for this will be evident by considering the characteristic current of an SPM machine, defined as

$$I_x \equiv \frac{\Psi_m}{L_d} \quad [\mathrm{A}] \tag{2}$$

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where Ψ_m is the rms flux linkage due to the PMs and L_d is the d-axis inductance (which is equal to the q-axis inductance for SPM machines). It is well known that optimal flux weakening, for both SPM and IPM machines, occurs when the characteristic current I_x equals the rated current I_R [41], [42]. The symmetrical three-phase short-circuit current will then be equal to the rated machine current. The inductance of SPM machines is, however, relatively low. Furthermore, if the flux linkage Ψ_m is reduced, this compromises the torque capability. As a result, the characteristic current tends to be significantly higher than the rated current, which severely limits the constant-power, flux-weakening operational range.

However, recent work on FSCW SPM machines has shown that these machines have high values of stator leakage inductance (both harmonic leakage and slot leakage) [1], which allows them to achieve a wide constant power speed range (CPSR). This type of machine has potential for demanding traction applications requiring wide CPSR, high-power density, and high efficiency.

This section will cover the key papers addressing the flux-weakening capability of SPM machines equipped with FSCW.

In [43], EL-Refaie and Jahns showed analytically that the optimal flux-weakening condition could be achieved in SPM equipped with FSCW. They provided FEA validation as well as a design method to achieve this condition.

In [44], EL-Refaie *et al.* provided experimental verification that the optimal flux-weakening condition can be met in SPM machines using FSCW. A 36-slot/30-pole prototype was built and tested. It was shown that a wide CPSR of ∼8:1 could be achieved while achieving a high efficiency above 90% over the entire speed range.

In [45], Cros *et al.* qualitatively indicated that fractional-slot concentrated windings have the potential of improving the flux-weakening capabilities of SPM machines. They did not provide any detailed analysis or a design technique to take advantage of this design technique.

In [46], Magnussen *et al.* presented a SPM design using concentrated windings with the claim of optimum flux weakening. Again, no analysis or design procedures were provided. In addition, there were some issues with their design including presence of radial forces because their choice of the slot/pole combination was not adequate. In addition, the rotor losses were excessive and were not accounted for during the design process.

In [47], Zhu *et al.* presented an 18-slot/12 (0.5 slot/pole/phase)-pole Halbach magnetized SPM machine. Optimum flux weakening was achieved by adjusting the tooth width and the slot width. In other words, this was achieved by adjusting the slot leakage inductance. No specific design procedure was provided. In addition, the performance of the traditional 0.5-slot/pole/phase winding configuration is usually significantly inferior to the performance of the distributed winding configuration due to the significantly lower winding factor.

In [48], EL-Refaie and Jahns investigated the scalability of FSCW SPM machines designed for wide CPSR. The study showed that this type of designs scale with number of poles, machine aspect ratio, and machine output power.

In [49], EL-Refaie *et al.* investigated the effect of having a back-EMF constraint on FSCW SPM machines designed for wide CPSR and specially targeting traction attractions. The study showed that the key impact is on the peak power current and there is a much lower impact on the machine power density.

In [50], EL-Refaie *et al.* proposed a modified vector control method to maximize the partial load efficiency of FSCW SPM machines designed for wide CPSR and again, this was mainly traction applications. It was shown that this new proposed method could achieve significantly higher partial load efficiency values (particularly at lower speeds) compared to the conventional maximum Torque/Amp method.

TABLE III
COMPARISON OF DL AND SL WINDINGS

	Double-Layer Windings	Single-Layer Windings
# of coil sides/slot	2	1
Mutual phase coupling	Significant phase-phase coupling through mutual slot leakage	Very low mutual phase coupling (recommended for fault-tolerance)
End turns	Shorter end turns	Longer end turns
Phase inductance	Lower phase inductance due to lower phase leakage inductance	Higher phase inductance due to higher phase leakage inductance 9better flux-weakening capability)
Rotor losses	Lower rotor losses mainly due to lower fundamental stator mmf space harmonic component	Higher rotor losses mainly due to higher fundamental stator mmf space harmonic component
Slot/pole combinations	Many slot/pole combinations can support DL windings	Few slot/pole combinations can support SL windings
Ease of manufacturing	More difficult to manufacture since there are 2 coil sides/slot	Easier to manufacture since there is 1 coil sides/slot
Synchronous winding factor	Lower	Higher
Back emf	More sinusoidal	Less sinusoidal

VI. SL VERSUS DL WINDINGS

There are two main types of FSCW as previously mentioned. SL stator winding configurations have stator slots that are each occupied by the coil sides of a single stator phase, while each slot in a DL winding configuration is split equally between coil sides from two phases (Fig. 2). There is a third type, which is FSCW with unequal tooth width that is not shown in the figure. This section will cover the key papers comparing the various types of FSCW mainly SL versus DL winding configurations. Table III provides a high-level summary of the key points of comparison between the two types of windings.

In [2], [51], Ishak *et al.* presented a comparison of SPM machines equipped with all teeth and alternative teeth (DL and SL) concentrated windings. It was shown that the alternative teeth (SL) winding configuration provides higher self-inductance and lower mutual inductance hence better fault-tolerance capability and flux-weakening capability. In addition, it was shown that SL winding has less sinusoidal back EMF due to the higher winding factors. Both winding configurations

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have similar cogging torque as they have the same slot/pole combination.

In [52], Ishak *et al.* presented a comparison of SPM machines equipped with concentrated windings (both SL and DL winding configurations) with equal tooth widths and SL concentrated windings with unequal teeth widths for the same slot/pole combination. It was shown that the SL winding configuration with unequal teeth widths provides higher torque capability and lower torque ripple.

In [53], EL-Refaie and Jahns investigated the impact of the number of winding layers as well as magnet type (sintered versus bonded) on the various performance aspects of FSCW SPM machines designed for wide CPSR targeting traction applications. The key conclusions were: 1) Sintered magnets offer opportunities for increasing the machine torque density compared to bonded-magnet designs, but their increased vulnerability to magnet eddy-current losses must be specifically addressed. 2) Machines with DL stator windings have lower spatial subharmonic components in their stator air-gap MMF distributions, resulting in lower torque ripple and magnet eddy-current losses compared to the SL winding designs. 3) However, machines with DL windings may suffer from lower overload torque capability compared to their SL winding counterparts due to higher magnetic saturation of the stator tooth tips in the DL winding machines.

VII. Rotor Losses

One of the key challenges of using FSCW configurations is the significant rotor losses (including magnet losses, rotor core losses, and sleeve losses in case of conductive sleeve) particularly at high speeds due to the various sub- and superspace-harmonic components inherent to such winding configurations that are not in synchronism with the rotor.

Several authors investigated the rotor losses with special focus on magnet losses. Effect of both circumferencial as well as axial segmentation has been investigated. The effect of various slot/pole combinations and number of phases on rotor losses has been investigated.

Losses in conductive retaining sleeves have been investigated. Effect of methods for reducing sleeve losses (including axial segmentation as well as copper cladding) has been investigated. This section will cover the key papers addressing the various aspects of rotor losses in PM synchronous machines using FSCW.

In [54], Atallah *et al.* presented an analytical model for calculating magnet losses. This model is powerful as it can account for the effect of peripheral segmentation of the magnets. The model has been used to evaluate examples of FSCW SPM machines. This model has been used in [55]–[57]. It was assumed that the stator current is pure sinusoidal.

In [55], it was used to calculate the losses for modular SPM designs that were introduced in [21].

In [56], [57], Ishak *et al.* used the model to compare the eddy current losses in the magnets for both SL and DL winding configurations for both BLDC and BLAC modes of operation. It was shown that the SL winding induces higher eddy current losses in the magnets due to the higher special harmonic content. In addition, it was shown that the induced losses are higher in case of BLDC operation due to the current time harmonic content. In addition, it was shown that the effect of magnet curvature on the losses is very small particularly in the case of high number of poles.

In [58], Nakano *et al.* used the model to compare the losses for various slot/pole combinations that can support concentrated windings.

In [59], Bianchi *et al.* proposed a general method for evaluating the rotor losses in three-phase fractional-slot PM machines. They studied the effect of the various slot/pole combinations on the rotor losses. A nonconductive retaining sleeve was assumed.

In [60], Polinder *et al.* presented a model for calculating eddy current losses in solid rotor back iron in FSCW PM machines. Even though the model results did not match well with experimental results, they concluded that the losses in a solid rotor for this type of machines would be unacceptable.

In [61], Ede *et al.* investigated the effect of axial segmentation on reducing the magnet losses. They proposed a method that is computationally faster compared to a full 3-D FEA.

In [62], Shah and EL-Refaie proposed a method for calculating losses in conducting retaining sleeves of FSCW SPM machines. They examined the effect of sleeve axial segmentation and copper cladding on reducing the sleeve losses. They evaluated the sleeve losses for various number of phases and slot/pole combinations.

In [7], as previously mentioned, EL-Refaie *et al.* examined the effect of number of phases on losses in conducting sleeves of FSCW SPM machines. A rotor loss FOM was introduced and evaluated for the various slot/pole combinations covering the feasible design space for four-, five-, and six-phase designs.

In [63], Ede *et al.* proposed an optimal torque control strategy for fault-tolerant PM brushless machines (equipped with FSCW). It has been shown that the adoption of a torque control strategy to minimize torque ripple under open-circuit and short-circuit fault conditions may lead to a significant increase in the eddy-current loss in the PMs of such machines.

VIII. Parasitic Effects

Parasitic effects such as noise, vibration, unbalanced magnetic forces, and torque ripple are always a concern when designing an electrical machines. These parasitic effects can potentially be higher in FSCW PM machines due to the additional harmonic contents This section is going to cover the key papers addressing the parasitic effects in FSCW PM machines with special focus on vibrations.

In [64], Chen *et al.* presented a method for predicting the electromagnetic vibration of PM brushless motors having a fractional number of slots per pole. The method has been validated experimentally. The method has been used to predict the vibration of PM brushless motors having different fractional slot/pole number combinations

In [65], [66], Wang *et al.* analyzed the radial force density harmonics and vibration characteristics of three-phase modular PM BLAC machines, in which the coils that belong to each phase are concentrated and wound on adjacent or alternative teeth, are. It is shown that, due to the presence of a large number

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of low- and high-order space-harmonic MMFs, it is more likely that low-frequency modes of vibration are excited in modular machines. Consequently, modular machines are more susceptible to low-frequency resonant vibrations. Experimental results validate the analysis and its findings.

In [67], Zhu *et al.* developed a general analytical model, formulated in 2-D polar coordinates, to predict the unbalanced magnetic force, which results in PM BLAC and BLDC machines having a diametrically asymmetric disposition of slots and phase windings. It is shown that the unbalanced magnetic force can be significant in machines having a fractional ratio of slot number to pole number, particularly when the electric loading is high. The developed model is validated by finite-element (FE) calculations on nine-slot/eight-pole and three-slot/two-pole machines. In addition, the unbalanced magnetic force has been measured on a prototype three-slot/two-pole machine and shown to be in excellent agreement with predicted results.

In [68], Magnussen and Lendenmann investigated the increase in parasitic effects in FSCW PM machines. This includes ripple torque, alternating magnetic fields in the rotor, unbalanced radial forces, and magnetic noise. This paper describes the reasons for the parasitic effects, in which machine topologies are particularly sensitive, and suggests measures in order to reduce their importance. Both traditional and modular concentrated windings are analyzed, as well as DL and SL windings.

Measurements on a prototype motor and three commercial servomotors have demonstrated that modular motors are favorable regarding ripple torque minimization.

IX. Fault Tolerance

Fault tolerance is one of the key issues with PM machines, in general, particularly in safety-critical applications. The main reason is that the PMs cannot be deexcited in case of a fault particularly in case of a generator that is coupled to a prime mover.

The key fault-tolerance requirements have been identified in literatures as follows:

1) complete electric isolation between phases;
2) implicit limiting of fault currents;
3) magnetic isolation between phases;
4) effective thermal isolation between phases;
5) physical isolation between phases;
6) higher number of phases.

These requirements can be met by using multiphase SL FSCW where each phase is fed by a single-phase H-bridge power converter. This section will cover the key papers addressing fault tolerance in PM synchronous machines using FSCW configurations.

In [69], Bianchi *et al.* presented some design considerations of fault-tolerant synchronous motors, characterized by a fractional number of slots per pole per phase. The first advantage of this configuration is a smooth torque, because of the elimination of the periodicity between slots and poles. The second one is a higher fault-tolerant capability making the machine able to work even in faulty conditions. However, the fractional-slot configuration presents high contents of MMF harmonics that may cause an unbalanced saturation and thus an unbearable torque ripple.

A method to design fractional-slot motors was illustrated in this paper, including DL and SL winding. The analytical computation is extended to determine the harmonics of MMF distribution. Their effect is highlighted in isotropic as well as anisotropic motors. Finally, some considerations are reported to avoid unsuitable configurations.

In [70], [71], Bianchi *et al.* presented postfault current control strategies of a five-phase PM motor. The analysis covers both the open circuit of one and two phases and the short circuit at the machine terminal of one phase. The proposed control guarantees safe drive operation after any fault occurrence. For the sake of generality, an analytical model has been used to investigate the properties of each postfault strategy. The results are general, and they apply to PM motor of any power rating. Simulations and experimental results validate the theoretical predictions.

In general, five-phase fault-tolerant PM machines received many attention in literature. Among its key advantages are high torque density, high controllability, reliability, and smooth torque production in case of a fault. Several papers addressed the various design aspects of five-phase fault-tolerant PM machines, power converter topologies as well as postfault current control strategies for different types of faults [72]–[74].

In [75], Chai *et al.* analyzed feasible slot and pole number combinations for multiplex two-phase and three-phase fault-tolerant PM machines and evaluated their relative merits via a design case study. An effective winding short-circuit detection technique based on search coils wound around the stator teeth was also presented, and its performance was assessed. It was shown that the proposed detection technique can reliably detect any type of short-circuit fault under all load conditions.

In [76], Mitcham *et al.* presented a new approach for selecting pole and slot numbers for fault-tolerant PM machines so that there is inherently negligible coupling between phases (regardless of other design detail). The preferred slot and pole number combinations thereby help to ensure that a fault in one phase does not undesirably affect the remaining unfaulted phases. Other well-known criteria for fault-tolerant operation, including high phase inductance, also have to be met. It was demonstrated how particular slot and pole combinations can be used to eliminate low pole number armature MMF, thereby reducing the vibration and stray loss present during normal operation.

In [77], Mecrow *et al.* examined the use of PM machine drives in high-performance safety-critical applications. Likely fault modes were identified and machine designs were developed for fault-tolerant operation, without severely compromising the drive performance. Fault tolerance was achieved by adopting a modular approach to the drive, with each phase electrically, magnetically, thermally, and physically independent of all others. Power converter requirements were discussed and methods for controlling a faulted phase developed to minimize the impact of a machine or power converter fault.

In [78], Haylock *et al.* discussed the design of a fault tolerant PM drive based on a 16-kW 13 000-r/min six-phase aircraft

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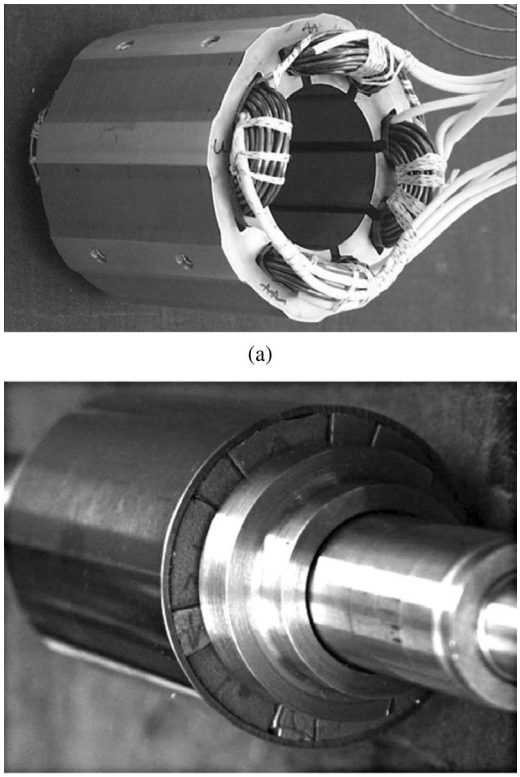

(a)

(b)

Fig. 6. (a) Four-phase stator. (b) Six-pole rotor Halbach array.

fuel pump specification. A "proof of concept" demonstrator has been built to this design and key parameters measured on the demonstrator drive are given. A novel current controller with near optimal transient performance was developed to enable precise shaping of the phase currents at high shaft speeds. A list of the most likely electrical faults was considered. Fault detection and identification schemes are developed for rapid detection of turn-to-turn faults and power device short circuit faults. Postfault control strategies were described which enable the drive to continue to operate indefinitely in the presence of each fault. Finally, results showed the initially healthy drive operating up to, through and beyond the introduction of two of the more serious faults.

In [79], Mecrow *et al.* discussed the design and testing of an aircraft electric fuel pump drive. The drive is a modular four-phase fault-tolerant system, which is designed to meet the specification with a fault in any one of the phases. The motor employed has a PM rotor with the magnets arranged in a Halbach array to maximize the air-gap flux density (Fig. 6). Exceptionally high electric loadings are obtained by flooding the entire motor with aircraft fuel, which acts as an excellent cooling agent. Theoretical results are compared with test results gained in conditions approaching those found in an aircraft. Tests are carried out on the unfaulted drive and with one of the several fault scenarios imposed. The electrical and thermal performance of the drive is assessed, showing how the flooded fuel cooling has excellent performance without introducing significant drag on the rotor.

In [80], Atkinson *et al.* discussed the design of a fault-tolerant electric motor for an aircraft main engine fuel pump. The motor in question is a four-phase fault-tolerant motor with separated windings and a six-pole PM rotor. Methods of reducing machine losses in both the rotor and stator were introduced and discussed. The methods used to calculate rotor eddy current losses were examined. Full 3-D FE time stepping, 2-D FE time stepping, and 2-D FE harmonic methods were discussed, and the differences between them and the results they produce were investigated. Conclusions were drawn about the accuracy of the results produced and how the methods in question will help the machine designer.

Since the machine phase inductance plays a key role in determining the machine fault currents as well as coupling between the various phases and hence fault tolerance, accurate calculation of the inductance during the design phase is critical. In [81]–[83], Zhu *et al.* presented an accurate method for calculating inductances of SPM machines with special focus on FSCW configurations.

TABLE IV
COMPARISON OF SPM VERSUS IPM MACHINES EQUIPPED WITH FSCW

	SPM	IPM
Stator structure	Can be segmented or continuous laminations	Can be segmented or continuous laminations
Magnet retention	Metallic or nonmetallic sleeves	Bridges and/or center posts
Flux-weakening	FSCW significantly improves flux-weakening capability	Has inherent good flux-weakening capability
Fault-tolerance	Improved fault-tolerance with FSCW	Improved fault-tolerance with FSCW
Rotor losses	Concentrated in retaining sleeve (if metallic) and permanent magnets	Concentrated in rotor laminations and areas of permanent magnets closest to air gap
Reluctance torque	Almost no reluctance torque	Reduced reluctance torque when equipped with FSCW compared to distributed windings

X. COMPARISON OF SPM VERSUS IPM

Even though most of the work done up to date focused on SPM machines, there is a growing interest in the use of IPM machines equipped with FSCW. The hope is that FSCW IPM machines will combine the benefits of the FSCW previously mentioned in addition to the benefits of an IPM rotor in terms of potentially reducing magnet content as well as easier magnet retention compared to SPM machines. This area is by no means mature, and much work is needed to fully understand the tradeoffs involved. This section will focus on the key papers that provided comparison between IPM and SPM machines with FSCW configurations. Table IV provides a high-level comparison of SPM versus IPM machines equipped with FSCW.

In [84], Bianchi *et al.* presented a comparison of fractional-slot SPM and IPM servomotors. Points of comparison included cogging torque, torque ripple, overload capability, and flux-weakening capability.

In [85], [86], Salminen *et al.* investigated the performance of fractional-slot concentrated windings for low-speed applications. They presented comparison of cogging torque, ripple torque, and back EMF for various slot/pole combinations for

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both surface and IPM designs. They concluded that smooth torque production is possible with such winding configurations.

In [87], Murakami *et al.* presented a comparative analysis and test results performed on three IPM machine designs using concentrated windings with different saliency ratios in order to identify the flux distribution and thus iron loss effects of using concentrated windings to determine the best rotor configuration for the next generation compressor motor for the 42-V car air conditioning. It was shown that using concentrated windings makes the motor more susceptible to iron losses at high speeds. Hence, in order to achieve high efficiency at high speeds, an IPM machine with low q-axis inductance is appropriate as it can reduce iron losses at high speeds. In addition, it was shown that using rectangular wires instead of conventional round wires reduces the end coil region by 15% and a higher slot fill factor can be achieved. It was shown that using an improved stator tooth configuration where the air gap is larger at the high stress points helps reducing the vibrations and noise in the machine.

In [88], Asano *et al.* presented a comparative analysis of vibration and noise production in SPM machines and IPM machines equipped with concentrated windings. A method for reducing the vibrations and noise was proposed by shaping the stator teeth in order to vary the air-gap thickness. It was shown that the noise in IPM machines is larger than in SPM machines. In addition, it was shown that the proposed noise reduction method is less effective in the case of IPM machines.

In [89], Cheng *et al.* presented a comparison between three various rotor structures for a starter/alternator used in a hybrid vehicle and equipped with concentrated windings in order to gain high torque density and high efficiency. The three rotor structures under consideration are surface mounted, inset radial, and inset tangential PM designs. It was shown the surface-mounted structure is the most effective in reducing the torque ripple it can increase the difficulty and cost of manufacturing compared to the two other designs.

In [90], Zhu *et al.* investigated the cogging reduction in IPM machines in both cases of full-pitch overlapping windings and short-pitch nonoverlapping concentrated windings. It was shown that by appropriately adjusting the pole-arc to pole-pitch ratio, the optimum ratio for cogging torque minimization that was derived for SPM machines is equally applicable in the case of IPM machines. It was also shown that the cogging torque in case of the nonoverlapping concentrated windings is almost half that in the case of full-pitch overlapping windings.

In [91], EL-Refaie and Jahns presented a detailed comparison of the high-speed operating characteristics of four synchronous PM machines for applications that require wide speed ranges of constant-power operation. These machines include SPM machines with both distributed and fractional-slot concentrated windings, and two IPM machines with distributed windings. These two versions of the IPM machine include one with and a tight constraint on the machine's back-EMF voltage at top speed and one without this constraint. The target application is an automotive direct-drive starter/alternator requiring a very wide 10 : 1 constant power speed ratio. Detailed comparisons of the performance characteristics of the machines were presented that include important issues such as the back-EMF voltage at top speed, machine mass and cost, and eddy current losses in the magnets. Analytical results were verified using FEA. Guidelines were developed to help drive designers decide which type of machine is most suitable for high-CPSR applications. Tradeoffs associated with choosing each of these machines are presented.

Fig. 7. Axial-flux machine made of SMC and Laminated back iron [93].

In [92], EL-Refaie and Jahns presented a thorough comparison of the converter performance characteristics of three types of synchronous PM machines presented in [91]. Detailed comparisons of the converter performance below and above the base speed were presented. Comparisons include important issues such as the converter switching and conduction losses, output ripple current, pulsewidth modulation copper and core losses, dc link current ripple, and bearing currents.

XI. Axial-Flux, Tubular, and Flux-Switching Machines

The main focus so far has been on radial-flux FSCW PM machines. There is a growing interest in other types of PM machines equipped with FSCW. Section XI will cover the key papers addressing the use of FSCW in other types of machines mainly axial-flux, tubular PM machines, and flux-switching machines.

In [93], Jack *et al.* described the design and construction of an axial-flux PM machine in which the teeth are manufactured from compacted insulated iron powder and the core back is formed from a strip-punched lamination formed, into a circle to grip the teeth (Fig. 7). This new method of construction overcomes the problems associated with punching and winding axial-flux machines formed using index-punched spirally wound laminations (the current state of the art). The construction has been shown to be mechanically stable and rather simple to manufacture. As well as production advantages it offers very high fill factor coils, which can significantly enhance the performance of this type of machine. In [94] and [95], the design and control of a fault-tolerant seven-phase axial-flux PM machine have been presented.

In [96], Caricchi *et al.* presented an innovative inverter topology for supplying an axial-flux PM machine using concentrated windings. This new topology permits shaping of the inverter output current waveform to be suitably adjusted with respect to the machine back-EMF waveform. This is done by adding a fourth leg or branch to the inverter devoted to controlling the voltage of the neutral point resulting for the star connection of the machine three phases. Improvements in the average torque production have been shown and verified experimentally.

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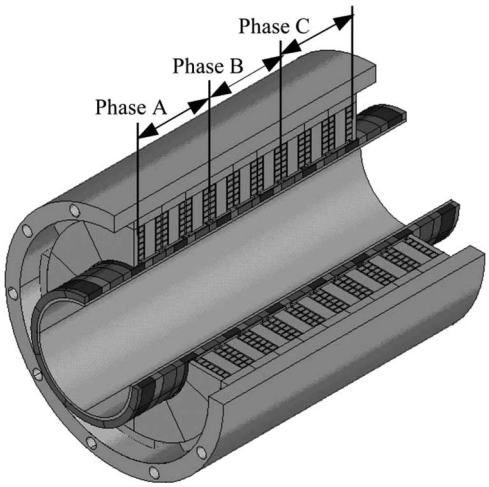

Fig. 8. Schematic of three-phase nine-slot/ten-pole tubular PM machine with quasi-Halbach magnetized armature [97].

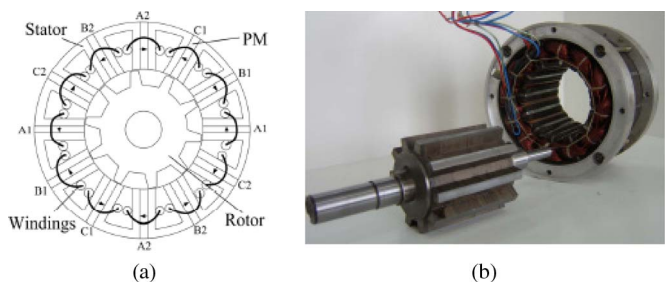

Fig. 9. (a) Cross section of a three-phase 12/10 FSPM [100]. (b) Prototype of a three-phase 12/10 FSPM [100].

In [97], Wang *et al.* discussed issues that are pertinent to the design of a linear PM generator for application in a free-piston energy converter. To achieve the required high-power density, high efficiency, and low moving mass, a tubular machine equipped with a modular stator winding and a quasi-Halbach magnetized armature is employed. It was shown that the machine design could be optimized with respect to three key dimensional ratios while satisfying other performance requirements. It was also shown that, when the generator is interfaced to an electrical system via a power electronic converter, both the converter volt-amps rating and the converter loss should be taken into account when optimizing the machine design. The performance of such a tubular generator is demonstrated by measurements on a ten-pole/nine-slot prototype machine (Fig. 8).

In [98] and [99], Amara *et al.* proposed analytical models for calculating the various loss components (both on the stator and rotor sides) of tubular modular PM machines.

Another family of machines that has been introduced in literature is the flux-switching/reversal PM (FSPM) machines (Fig. 9). At a high level, this type of machines is comparable to Doubly Salient PM machines [100]. They have been investigated for various applications [101]–[103]. Fault-tolerant flux-switching PM machines have been investigated. The same concepts previously covered in can be applied to this type of machines including the multiphase approach [104], [105], as shown in Figs. 10 and 11.

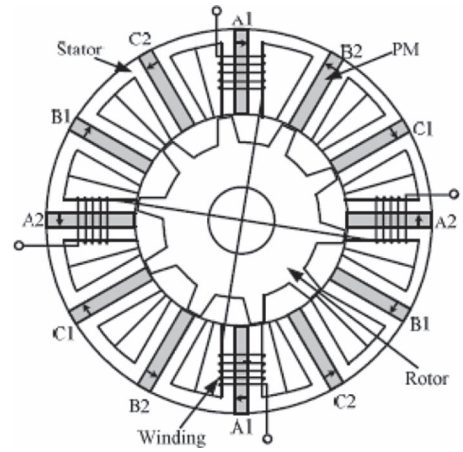

Fig. 10. Cross section of a redundant FSPM [104].

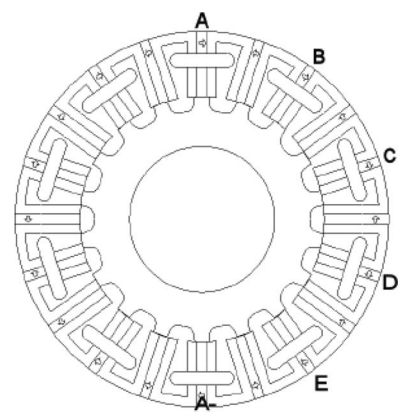

Fig. 11. Cross section of a five-phase alternate-pole wound FSPM [105].

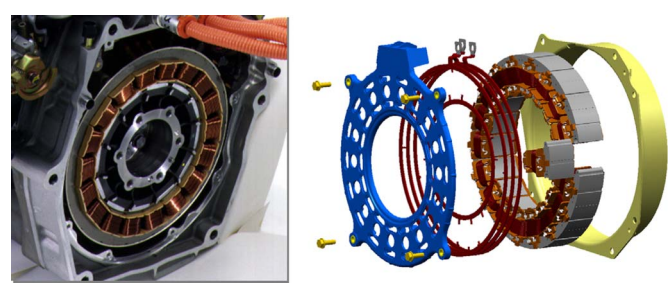

Fig. 12. Stator of the Honda Insight.

XII. Commercial Applications and Future Evolution of Research

FSCW have already been used in commercial PM machines that are used in various applications. This is mainly due to the various advantages covered in the previous sections. This section will provide a summary of the commercial applications where FSCW PM machines. Several examples that already exist in the public domain will be presented. In addition, some applications where FSCW PM machines are being investigated for potential future use will be discussed.

The first example is the segmented stator structures with FSCW of the various Honda designs. These designs have a significantly high slot fill factor. They use the more traditional 0.5 slot/pole/phase, which has relatively low winding factor of 0.866. Fig. 12 shows the stator of the Honda Insight, which

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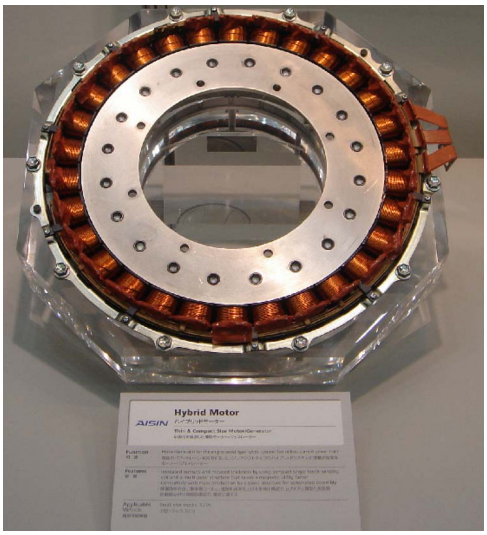

Fig. 13. Toyota/Aisin Motor.

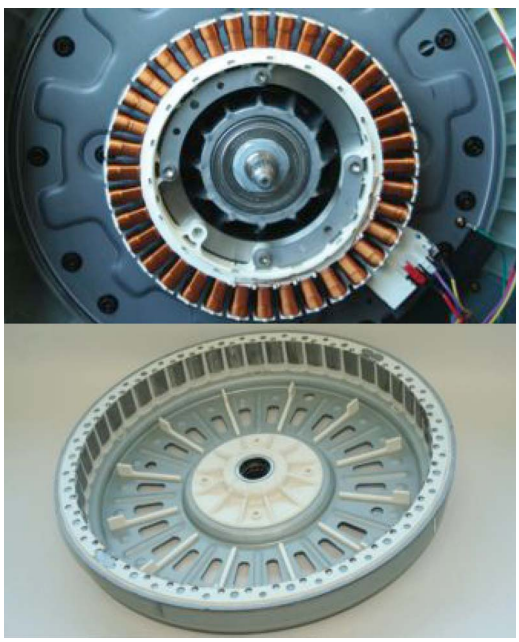

Fig. 14. Whirlpool inside/out PM machine for washing machine application.

is very similar to the stator of the Honda Accord. The Honda Insight machine is an SPM while the Accord machine is an IPM.

Another example is the starter/alternator that was developed by ZF Sachs [32], and shown in Fig. 4. This design has a plug-in-tooth segmented stator structure as previously mentioned. This design is equipped with 2/7 slot/pole/phase, which has high winding factor (0.933 for DL, and 0.966 for SL). The key challenge with this winding configuration is the high rotor losses due to the dominant sub- and superspace-harmonic components.

Fig. 13 shows the stator of a Toyota/Aisin FSCW PM traction motor that was exhibited at the Electric Vehicle Symposium 22 in Yokohama, Japan.

Fig. 14 shows an inside/out FSCW PM machine by Whirlpool that is used in a washing machine application.

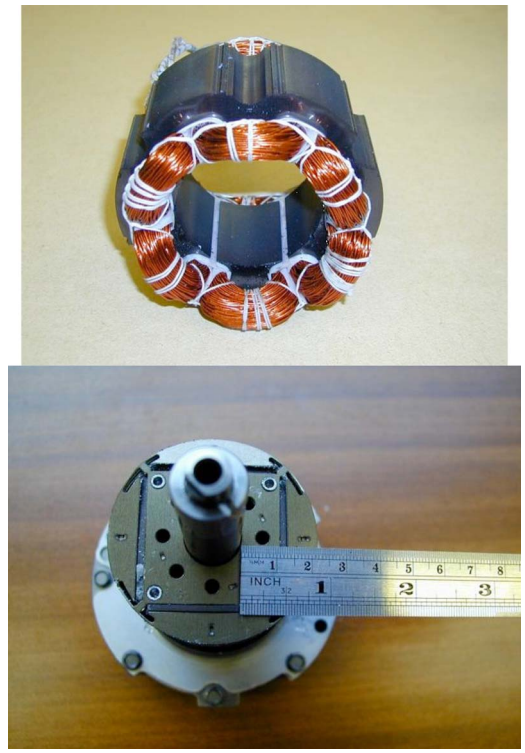

Fig. 15. FSCW IPM machine by Panasonic.

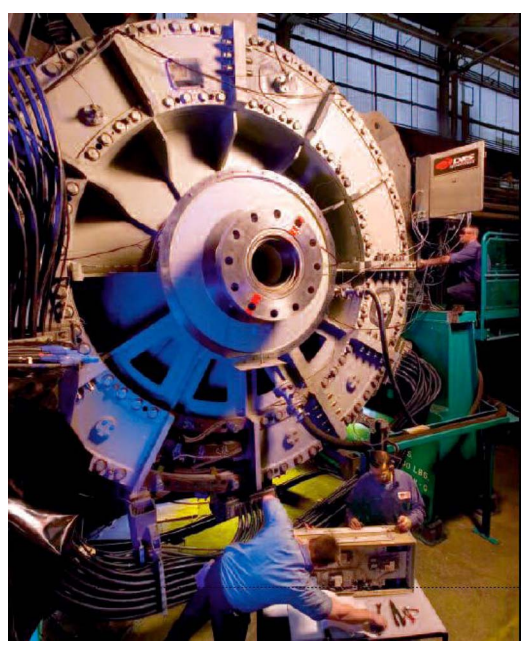

Fig. 16. Eighteen-megawatt PM ship propulsion machine by DRS.

Fig. 15 shows an FSCW IPM air-conditioner scroll compressor pump motor by Panasonic.

FSCW PM machines are also being considered and evaluated for a wide range of applications including ship propulsion (Fig. 16), wind generators (Fig. 17), ocean wave generators, and aerospace applications. Even though the examples in Figs. 16 and 17 might not be using FSCW configurations, they give a good idea of the machine sizes for which FSCW configurations are being considered.

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Fig. 17. Five-megawatt 147-r/min PM wind generator by Prokon Nord.

FSCW PM machines are particularly a good fit for marine propulsion and wind applications. Since these are high-torque low-speed applications, they lend themselves to a higher number of poles, which is suitable for FSCW. In addition, since they are low-speed applications, the higher rotor losses due to FSCW are more manageable compared to high-speed applications.

XIII. Conclusion

This paper provides a thorough comprehensive analysis of the tradeoffs involved in using FSCW in PM synchronous machines. The key topics that are covered include the following: the theory behind FSCW PM machines, design and analysis of FSCW PM machines, achieving high-power density, SL versus DL windings, rotor losses, fault tolerance, parasitic effects, SPM versus IPM, axial-flux, tubular, and flux-switching PM machines.

A summary of commercial applications where FSCW PM are used has been presented, and several examples were included. In addition, the key applications where FSCW PM machines are being considered for potential future use have been discussed.

This paper will provide engineers and researchers interested in the area of synchronous PM machines equipped with FSCW a comprehensive reference that will help them come up to speed with what has been done in this fast-growing area.

References

[1] A. M. EL-Refaie, Z. Q. Zhu, T. M. Jahns, and D. Howe, "Winding inductances of fractional slot surface-mounted permanent magnet brushless machines," in *Conf. Rec. IEEE IAS Annu. Meeting*, Edmonton, AB, Canada, Oct. 2–5, 2008, pp. 1–8.

[2] D. Ishak, Z. Q. Zhu, and D. Howe, "Comparison of PM brushless motors, having either all teeth or alternate teeth wound," *IEEE Trans. Energy Convers.*, vol. 21, no. 1, pp. 95–103, Mar. 2006.

[3] J. Cros and P. Viarouge, "Synthesis of high performance PM motors with concentrated windings," *IEEE Trans. Energy Convers.*, vol. 17, no. 2, pp. 248–253, Jun. 2002.

[4] F. Magnussen and C. Sadarangani, "Winding factors and Joule losses of permanent magnet machines with concentrated windings," in *Proc. IEEE-IEMDC*, Madison, WI, Jun. 2003, vol. 1, pp. 333–339.

[5] N. Bianchi, S. Bolognani, and G. Grezzani, "Design considerations for fractional-slot winding configurations of synchronous machines," *IEEE Trans. Ind. Appl.*, vol. 42, no. 4, pp. 997–1006, Jul./Aug. 2006.

[6] N. Bianchi and M. Dai Pre, "Use of the star of slots in designing fractional-slot single-layer synchronous motors," *Proc. Inst. Elect. Eng.—Electr. Power Appl.*, vol. 153, no. 3, pp. 997–1006, May 2006.

[7] A. M. EL-Refaie, M. R. Shah, R. Qu, and J. M. Kern, "Effect of number of phases on losses in conducting sleeves of high speed surface PM machine rotors," in *Conf. Rec. 42nd IEEE IAS Annu. Meeting*, New Orleans, LA, Sep. 2007, pp. 1522–1529.

[8] T. Katsuma and M. Kitoh, "Brushless motor having permanent magnet rotor and salient pole stator," U.S. Patent 4 719 378, Jan. 12, 1988.

[9] K. F. Konecny, "Compact three-phase permanent magnet rotary machine having low vibration and high performance," U.S. Patent 4 774 428, Sep. 27, 1988.

[10] S. Nishio, "Polyphase direct current motor," U.S. Patent 5 006 745, Apr. 9, 1991.

[11] A. G. Jack, B. C. Mecrow, P. G. Dickinson, D. Stephenson, J. S. Burdess, N. Fawcett, and J. T. Evans, "Permanent magnet machines with powdered iron cores and pressed windings," *IEEE Trans. Ind. Appl.*, vol. 36, no. 4, pp. 1077–1084, Jul./Aug. 2000.

[12] B. Huang and A. Hartman, "High speed ten pole/twelve slot DC brushless motor with minimized net radial force and low cogging torque," U.S. Patent 5 675 196, Oct. 7, 1997.

[13] R. Dhawan and Z. Soghomonian, "Seven phase brush-less synchronous motor with reduced inverter size," in *Proc. APEC*, Feb. 2004, vol. 2, pp. 1099–1105.

[14] A. V. Pyntikov and B. A. Maslov, "Multiple magnetic path electric motor," U.S. Patent 6 384 496, May 7, 2002.

[15] B. Maslov and A. Pyntikov, "Rotary electric motor having magnetically isolated stator and rotor groups," U.S. Patent 6 492 756, Dec. 10, 2002.

[16] B. A. Maslov and Z. Soghomonian, "Rotary electric motor having axially aligned stator poles and/or rotor poles," U.S. Patent 6 617 746, Sep. 9, 2003.

[17] B. A. Maslov and Z. Soghomonian, "Rotary electric motor having at least three axially aligned stator poles and/or rotor poles," U.S. Patent 6 710 502, Mar. 23, 2004.

[18] Z. Soghomonian, B. A. Maslov, and M. A. Benson, "Rotary electric motor having a plurality of skewed stator poles and/or rotor poles," U.S. Patent 6 717 323, Apr. 6, 2004.

[19] F. Libert and J. Soulard, "Investigation on pole-slot combinations for permanent-magnet machines with concentrated windings," in *Proc. ICEM*, 2004, pp. 530–535.

[20] K. Reichert, "PM-motors with concentrated, non-overlapping windings, some characteristics," in *Proc. ICEM*, 2004, p. 541.

[21] Z. P. Xia, J. Wang, and D. Howe, "Three-phase interior magnet modular brushless machines for automotive applications," in *Proc. ICEM*, 2004.

[22] K. Atallah, J. Wang, and D. Howe, "Torque-ripple minimization in modular permanent-magnet brushless machines," *IEEE Trans. Ind. Appl.*, vol. 39, no. 6, pp. 1689–1695, Nov./Dec. 2003.

[23] J. Wang, Z. P. Xia, and D. Howe, "Analysis of three-phase surface-mounted magnet modular permanent magnet machines," in *Proc. 2nd IEE Int. Conf. PEMD*, Mar./Apr. 2004, vol. 3, pp. 27–32.

[24] A. Di Gerlando, R. Perini, and M. Ubaldini, "High pole number, PM synchronous motor with concentrated coil armature windings," in *Proc. ICEM*, 2004.

[25] C. Noel, N. Takorabet, and F. Meibody-Tabar, "Short-circuit current reduction technique for surface mounted PM machines," in *Conf. Rec. IEEE IAS Annu. Meeting*, Oct. 2004, vol. 3, pp. 1427–1433.

[26] F. Tajima, "Permanent magnet rotating electric machine and electrically driven vehicle employing same," U.S. Patent 6 396 183, May 28, 2002.

[27] M. Kimura, "Permanent magnet rotating electric machine," U.S. Patent 6 462 451, Oct. 8, 2002.

[28] H. Koharagi, "Permanent magnet rotating electric machine," U.S. Patent 6 525 442, Feb. 25, 2003.

[29] R. Takahata, "Permanent magnet rotating electric machine," U.S. Patent 6 552 461, Apr. 22, 2003.

[30] Y. Asano, "Motor with permanent magnet," U.S. Patent 6 583 531, Jun. 24, 2003.

[31] M. Seguchi, "Rotary electric machine having a permanent magnet stator and permanent magnet rotor," U.S. Patent 6 590 312, Jul. 8, 2003.

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[32] N. Bianchi, M. Dai Pre, L. Alberti, and E. Fornasiero, "Theory and design of fractional-slot PM machines," in *Conf. Rec. IEEE IAS Annu. Meeting*, New Orleans, LA, Sep. 23, 2007.
[33] A. M. EL-Refaie, T. M. Jahns, and D. W. Novotny, "Analysis of surface permanent magnet machines equipped with concentrated windings," *IEEE Trans. Energy Convers.*, vol. 21, no. 1, pp. 34–43, Mar. 2006.
[34] [Online]. Available: www.zfsachs.com/dynastart-pc
[35] H. Akita, Y. Nakahara, N. Miyake, and T. Oikawa, "New core structure and manufacturing method for high efficiency of permanent magnet motors," in *Conf. Rec. IEEE IAS Annu. Meeting*, Salt Lake City, UT, Oct. 2003, vol. 2, pp. 367–372.
[36] J. Cros, P. Viarouge, and C. Gelinas, "Design of PM brushless motors using iron-resin composites for automotive applications," in *Conf. Rec. IEEE IAS Annu. Meeting*, Oct. 1998, vol. 1, pp. 5–11.
[37] J. Cros, P. Viarouge, and A. Halila, "Brush DC motors with concentrated windings and soft magnetic composites," in *Conf. Rec. IEEE IAS Annu. Meeting*, Oct. 2001, vol. 4, pp. 2549–2556.
[38] J. Cros, P. Viarouge, Y. Chalifour, and J. Figueroa, "A new structure of universal motor using soft magnetic composites," *IEEE Trans. Ind. Appl.*, vol. 40, no. 2, pp. 550–557, Mar./Apr. 2004.
[39] J. Cros and P. Viarouge, "New structures of polyphase claw-pole machines," *IEEE Trans. Ind. Appl.*, vol. 40, no. 1, pp. 113–120, Jan./Feb. 2004.
[40] J. Cros, P. Viarouge, R. Carlson, and L. V. Dokonal, "Comparison of brushless DC motors with concentrated winding and segmented stator," in *Proc. ICEM*, 2004, p. 761.
[41] W. Soong and T. J. E. Miller, "Field weakening performance of brushless synchronous AC motor drives," *Proc. Inst. Elect. Eng.—Electr. Power Appl.*, vol. 141, no. 6, pp. 331–340, Nov. 1994.
[42] R. F. Schiferl and T. A. Lipo, "Power capability of salient pole permanent magnet synchronous motor in variable speed drive applications," *IEEE Trans. Ind. Appl.*, vol. 26, no. 1, pp. 115–123, Jan./Feb. 1990.
[43] A. M. EL-Refaie and T. M. Jahns, "Optimal flux weakening in surface PM machines using concentrated windings," *IEEE Trans. Ind. Appl.*, vol. 41, no. 3, pp. 790–800, May/Jun. 2005.
[44] A. M. EL-Refaie, T. M. Jahns, P. J. McCleer, and J. W. McKeever, "Experimental verification of optimal flux weakening in surface PM machines using concentrated windings," *IEEE Trans. Ind. Appl.*, vol. 42, no. 2, pp. 443–453, Mar./Apr. 2006.
[45] J. Cros, J. R. Figueroa, and P. Viarouge, "BLDC motors with surface mounted PM rotor for wide constant power operation," in *Conf. Rec. IEEE IAS Annu. Meeting*, Oct. 2003, vol. 3, pp. 1933–1940.
[46] F. Magnussen, P. Thelin, and C. Sadarangani, "Performance evaluation of permanent magnet synchronous machines with concentrated and distributed windings including the effect of field weakening," in *Proc. 2nd IEE Int. Conf. PEMD*, Mar./Apr. 2004, vol. 2, pp. 679–685.
[47] Z. Q. Zhu, Z. P. Xia, Y. F. Shi, D. Howe, A. Pride, and X. J. Chen, "Performance of Halbach magnetized brushless AC motors," *IEEE Trans. Magn.*, vol. 39, no. 5, pp. 2992–2994, Sep. 2003.
[48] A. M. EL-Refaie and T. M. Jahns, "Scalability of surface PM machines with concentrated windings designed to achieve wide speed ranges of constant-power operation," *IEEE Trans. Energy Convers.*, vol. 21, no. 2, pp. 362–369, Jun. 2006.
[49] A. M. EL-Refaie, T. M. Jahns, and J. W. McKeever, "Effect of back-EMF constraints on fractional-slot surface PM machines designed for wide constant-power speed range operation," *Electr. Power Compon. Syst.*, vol. 36, no. 2, pp. 124–137, Feb. 2008.
[50] A. M. EL-Refaie, T. M. Jahns, and J. McKeever, "Modified vector control algorithm for increasing partial-load efficiency of fractional-slot concentrated winding surface PM machines," in *Conf. Rec. IEEE IAS Annu. Meeting*, Tampa, FL, Oct. 8–12, 2006, pp. 50–57.
[51] D. Ishak, Z. Q. Zhu, and D. Howe, "Comparative study of permanent magnet brushless motors with all teeth and alternative teeth windings," in *Proc. 2nd IEE Int. Conf. PEMD*, Mar./Apr. 2004, vol. 2, pp. 834–839.
[52] D. Ishak, Z. Q. Zhu, and D. Howe, "Permanent magnet brushless machines with unequal tooth widths and similar slot and pole numbers," *IEEE Trans. Ind. Appl.*, vol. 41, no. 2, pp. 584–590, Mar./Apr. 2005.
[53] A. M. EL-Refaie and T. M. Jahns, "Impact of winding layer number and magnet type on synchronous surface PM machines designed for wide constant-power speed range operation," in *Conf. Rec. IEEE IAS Annu. Meeting*, Tampa, FL, Oct. 8–12, 2006, pp. 1486–1493.
[54] K. Atallah, D. Howe, P. H. Mellor, and D. A. Stone, "Rotor loss in permanent-magnet brushless AC machines," *IEEE Trans. Ind. Appl.*, vol. 36, no. 6, pp. 1612–1618, Nov./Dec. 2000.
[55] H. Toda, Z. Xia, J. Wang, K. Atallah, and D. Howe, "Rotor eddy-current loss in permanent magnet brushless machines," *IEEE Trans. Magn.*, vol. 40, no. 4, pp. 2104–2106, Jul. 2004.
[56] D. Ishak, Z. Q. Zhu, and D. Howe, "Analytical prediction of rotor eddy current losses in permanent magnet brushless machines with all teeth and alternate teeth windings—Part I: Polar co-ordinate model," in *Proc. 7th ICEMS*, 2004.
[57] D. Ishak, Z. Q. Zhu, and D. Howe, "Analytical prediction of rotor eddy current losses in permanent magnet brushless machines with all teeth and alternate teeth windings—Part II: Rectangular co-ordinate model," in *Proc. 7th ICEMS*, 2004.
[58] M. Nakano, H. Kometani, and M. Kawamura, "A study on eddy-current losses in rotors of surface permanent magnet synchronous machines," in *Conf. Rec. IEEE IAS Annu. Meeting*, Oct. 2004, vol. 3, pp. 1696–1702.
[59] N. Bianchi, S. Bolognani, and E. Fornasiero, "A general approach to determine the rotor losses in three-phase fractional-slot PM machines," in *Proc. IEEE IEMDC*, 2007, vol. 1, pp. 634–641.
[60] H. Polinder, M. J. Hoeijmakers, and M. Scuotto, "Eddy-current losses in the solid back-iron of permanent-magnet machines with concentrated fractional pitch windings," in *Proc. 3rd IET Int. Conf. Power Electron., Mach., Drives*, Mar. 2006, pp. 479–483.
[61] J. D. Ede, K. Atallah, G. W. Jewell, J. B. Wang, and D. Howe, "Effect of axial segmentation of permanent magnets on rotor loss of modular brushless machines," in *Conf. Rec. IEEE IAS Annu. Meeting*, Oct. 2004, vol. 3, pp. 1703–1708.
[62] M. R. Shah and A. M. EL-Refaie, "Eddy current loss minimization in conducting sleeves of high speed machine rotors by optimal axial segmentation and copper cladding," in *Conf. Rec. 42nd IEEE IAS Annu. Meeting*, New Orleans, LA, Sep. 2007, pp. 544–551.
[63] J. D. Ede, K. Atallah, J. Wang, and D. Howe, "Effect of optimal torque control on rotor loss of fault-tolerant permanent-magnet brushless machines," *IEEE Trans. Magn.*, vol. 38, no. 5, pp. 3291–3293, Sep. 2002.
[64] Y. S. Chen, Z. Q. Zhu, and D. Howe, "Vibration of PM brushless machines having a fractional number of slots per pole," *IEEE Trans. Magn.*, vol. 42, no. 10, pp. 3395–3397, Oct. 2006.
[65] J. Wang, Z. P. Xia, D. Howe, and S. A. Long, "Vibration characteristics of modular permanent magnet brushless AC machines," in *Conf. Rec. 41st IEEE IAS Annu. Meeting*, Oct. 8–12, 2006, vol. 3, pp. 1501–1506.
[66] J. Wang, Z. P. Xia, S. A. Long, and D. Howe, "Radial force density and vibration characteristics of modular permanent magnet brushless AC machine," *Proc. Inst. Elect. Eng.—Electr. Power Appl.*, vol. 153, no. 6, pp. 793–801, Nov. 2006.
[67] Z. Q. Zhu, D. Ishak, D. Howe, and J. Chen, "Unbalanced magnetic forces in permanent-magnet brushless machines with diametrically asymmetric phase windings," *IEEE Trans. Ind. Appl.*, vol. 43, no. 6, pp. 1544–1553, Nov./Dec. 2007.
[68] F. Magnussen and H. Lendenmann, "Parasitic effects in PM machines with concentrated windings," *IEEE Trans. Ind. Appl.*, vol. 43, no. 5, pp. 1223–1232, Sep./Oct. 2007.
[69] N. Bianchi, M. Dai Pre, G. Grezzani, and S. Bolognani, "Design considerations on fractional-slot fault-tolerant synchronous motors," in *Proc. IEEE IEMDC*, San Antonio, TX, 2005, pp. 902–909.
[70] N. Bianchi, S. Bolognani, and M. D. Pre, "Strategies for the fault-tolerant current control of a five-phase permanent-magnet motor," *IEEE Trans. Ind. Appl.*, vol. 43, no. 4, pp. 960–970, Jul./Aug. 2007.
[71] N. Bianchi, S. Bolognani, and M. D. Pre, "Design and tests of a fault-tolerant five-phase permanent magnet motor," in *Proc. 37th IEEE PESC*, Jun. 18–22, 2006, pp. 1–8.
[72] N. Bianchi, S. Bolognani, and M. D. Pre, "Impact of stator winding of a five-phase motor on postfault operations," *IEEE Trans. Ind. Electron.*, vol. 55, no. 5, pp. 1978–1987, May 2008.
[73] C. Bianchini, E. Fornasiero, T. Matzen, N. Bianchi, and A. Bellini, "Fault detection of a five-phase permanent-magnet machine," in *Proc. 34th IEEE IECON*, Nov. 10–13, 2008, pp. 1200–1205.
[74] S. Dwari and L. Parsa, "An optimal control technique for multiphase PM machines under open-circuit faults," *IEEE Trans. Ind. Electron.*, vol. 55, no. 5, pp. 1988–1995, May 2008.
[75] J. Chai, J. Wang, K. Atallah, and D. Howe, "Performance comparison and winding fault detection of duplex 2-phase and 3-phase fault-tolerant permanent magnet brushless machines," in *Conf. Rec. 42nd IEEE IAS Annu. Meeting*, Sep. 23–27, 2007, pp. 566–572.
[76] A. J. Mitcham, A. G. Jack, J. A. Haylock, and J. Coles, "Favourable slot and pole combinations for fault-tolerant PM machines," *Proc. Inst. Elect. Eng.—Electr. Power Appl.*, vol. 151, no. 5, pp. 520–525, Sep. 2004.
[77] B. C. Mecrow, A. G. Jack, J. A. Haylock, and J. Coles, "Fault-tolerant permanent magnet machine drives," *Proc. Inst. Elect. Eng.—Electr. Power Appl.*, vol. 143, no. 6, pp. 437–442, Nov. 1996.
[78] J. A. Haylock, B. C. Mecrow, A. G. Jack, J. A. Haylock, and D. J. Atkinson, "Operation of a fault tolerant PM drive for an aerospace fuel

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pump application," *Proc. Inst. Elect. Eng.—Electr. Power Appl.*, vol. 145, no. 5, pp. 441–448, Sep. 1998.

[79] B. C. Mecrow, A. G. Jack, D. J. Atkinson, S. R. Green, G. J. Atkinson, A. King, and B. Green, "Design and testing of a four-phase fault-tolerant permanent-magnet machine for an engine fuel pump," *IEEE Trans. Energy Convers.*, vol. 19, no. 4, pp. 671–678, Dec. 2004.

[80] G. J. Atkinson, B. C. Mecrow, A. G. Jack, D. J. Atkinson, P. Sangha, and M. Benarous, "The analysis of losses in high-power fault-tolerant machines for aerospace applications," *IEEE Trans. Ind. Appl.*, vol. 42, no. 5, pp. 1162–1170, Sep./Oct. 2006.

[81] Z. Q. Zhu, D. Howe, and T. S. Birch, "Calculation of winding inductances of brushless motors with surface-mounted permanent magnets," in *Proc. ICEM*, Paris, France, 1994, pp. 327–332.

[82] Z. Q. Zhu and D. Howe, "Winding inductances of brushless machines with surface-mounted magnets," in *Proc. Int. Electr. Mach. Drives Conf.*, May 1997, pp. WB2/2.1–WB2/2.3.

[83] Z. Q. Zhu, D. Howe, and J. K. Mitchell, "Magnetic field analysis and inductances of brushless DC machines with surface-mounted magnets and non-overlapping stator windings," *IEEE Trans. Magn.*, vol. 31, no. 3, pp. 2115–2118, May 1995.

[84] N. Bianchi, S. Bolognani, and G. Grezzani, "Fractional-slot IPM servomotors: Analysis and performance comparisons," in *Proc. ICEM*, 2004, pp. 1–6.

[85] P. Salminen, J. Mantere, J. Pyrhonen, and M. Niemela, "Performance of fractional slot wound PM-motors," in *Proc. ICEM*, 2004.

[86] P. Salminen, M. Niemela, J. Pyrhonen, and J. Mantere, "Performance analysis of fractional slot wound PM-motors for low speed applications," in *Conf. Rec. IEEE IAS Annu. Meeting*, Oct. 2004, vol. 2, pp. 1032–1037.

[87] H. Murakami, H. Kataoka, Y. Honda, S. Morimoto, and Y. Takeda, "Highly efficient brushless motor design for an air-conditioner of the next generation 42 V vehicle," in *Conf. Rec. IEEE IAS Annu. Meeting*, Oct. 2001, vol. 1, pp. 461–466.

[88] Y. Asano, Y. Honda, H. Murakami, Y. Takeda, and S. Morimoto, "Novel noise improvement technique for a PMSM with concentrated winding," in *Proc. PCC*, Osaka, Japan, Apr. 2002, vol. 2, pp. 460–465.

[89] S. Cheng, Y. Pei, F. Chai, and S. Cui, "The torque pulsation analysis of a starter generator with concentrated windings based hybrid vehicles," in *Proc. IEEE-IEMDC*, Madison, WI, Jun. 2003, vol. 1, pp. 218–221.

[90] Z. Q. Zhu, S. Ruangsinchaiwanich, N. Schofield, and D. Howe, "Reduction of cogging torque in interior-magnet brushless machines," *IEEE Trans. Magn.*, vol. 39, no. 5, pp. 3238–3240, Sep. 2003.

[91] A. M. EL-Refaie and T. M. Jahns, "Comparison of synchronous PM machine types for wide constant-power speed range operation," in *Conf. Rec. IEEE IAS Annu. Meeting*, Hong Kong, Oct. 2–6, 2005, vol. 2, pp. 1015–1022.

[92] A. M. EL-Refaie and T. M. Jahns, "Comparison of synchronous PM machine types for wide constant-power speed operation: Converter performance," *IET Electr. Power Appl.*, vol. 1, no. 2, pp. 217–222, Mar. 2007.

[93] A. G. Jack, B. C. Mecrow, G. Nord, and P. G. Dickinson, "Axial flux motors using compacted insulated iron powder and laminations: Design and test results," in *Proc. IEEE IEMDC*, May 15–18, 2005, pp. 378–385.

[94] F. Locment, E. Semail, and X. Kestelyn, "Vectorial approach-based control of a seven-phase axial flux machine designed for fault operation," *IEEE Trans. Ind. Electron.*, vol. 55, no. 10, pp. 3682–3691, Oct. 2008.

[95] F. Locment, E. Semail, X. Kestelyn, and A. Bouscayrol, "Control of a seven-phase axial flux machine designed for fault operation," in *Proc. 32nd IEEE IECON*, Nov. 6–10, 2006, pp. 1101–1107.

[96] F. Caricchi, F. Crescimbini, T. A. Lipo, and E. Santini, "Innovative inverter topology for concentrated winding PM motor drives," in *Proc. IEEE Power Electron. Spec. Conf.*, Jun. 1992, vol. 2, pp. 964–972.

[97] J. Wang, M. West, D. Howe, H. Z. De La Parra, and W. M. Arshad, "Design and experimental verification of a linear permanent magnet generator for a free-piston energy converter," *IEEE Trans. Energy Convers.*, vol. 22, no. 2, pp. 299–306, Jun. 2007.

[98] Y. Amara, J. Wang, and D. Howe, "Analytical prediction of eddy-current loss in modular tubular permanent magnet machines," *IEEE Trans. Energy Convers.*, vol. 20, no. 4, pp. 761–770, Dec. 2005.

[99] Y. Amara, J. Wang, and D. Howe, "Stator iron loss of tubular permanent magnet machines," *IEEE Trans. Ind. Appl.*, vol. 41, no. 4, pp. 989–995, Jul./Aug. 2005.

[100] W. Hua, M. Cheng, H. Jia, and X. Fu, "Comparative study of flux-switching and doubly-salient PM machines particularly on torque capability," in *Conf. Rec. IEEE IAS Annu. Meeting*, Oct. 2008, vol. 1, pp. 1–8.

[101] Y. Cheng, C. Pollock, and H. Pollock, "A permanent magnet flux switching motor for low energy axial fans," in *Conf. Rec. IEEE IAS Annu. Meeting*, Oct. 2005, vol. 3, pp. 2168–2175.

[102] H. Pollock, C. Pollock, R. T. Walter, and B. V. Gorti, "Low cost, high power density, flux switching machines and drives for power tools," in *Conf. Rec. IEEE IAS Annu. Meeting*, Oct. 2003, vol. 3, pp. 1451–1457.

[103] C. Pollock, H. Pollock, R. Barron, R. Sutton, J. Coles, D. Moule, and A. Court, "Flux switching motors for automotive applications," in *Conf. Rec. IEEE IAS Annu. Meeting*, Oct. 2005, vol. 1, pp. 242–249.

[104] W. Zhao, M. Cheng, W. Hua, and H. Jia, "A redundant flux-switching permanent magnet motor drive for fault-tolerant applications," in *Proc. Vehicle Power Propulsion Conf.*, Sep. 2008, vol. 1, pp. 1–6.

[105] A. S. Thomas, Z. Q. Zhu, R. L. Owen, G. W. Jewell, and D. Howe, "Multi-phase flux-switching permanent magnet brushless machine for aerospace application," in *Conf. Rec. IEEE IAS Annu. Meeting*, Oct. 2005, vol. 1, pp. 1–8.

[106] M. Kaizuka, T. Imai, S. Ishikawa, M. Niki, and H. Adachi, "Development of 2005 model year accord hybrid," in *Proc. EVS21*, Monte Carlo, Monaco, Apr. 2005.

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